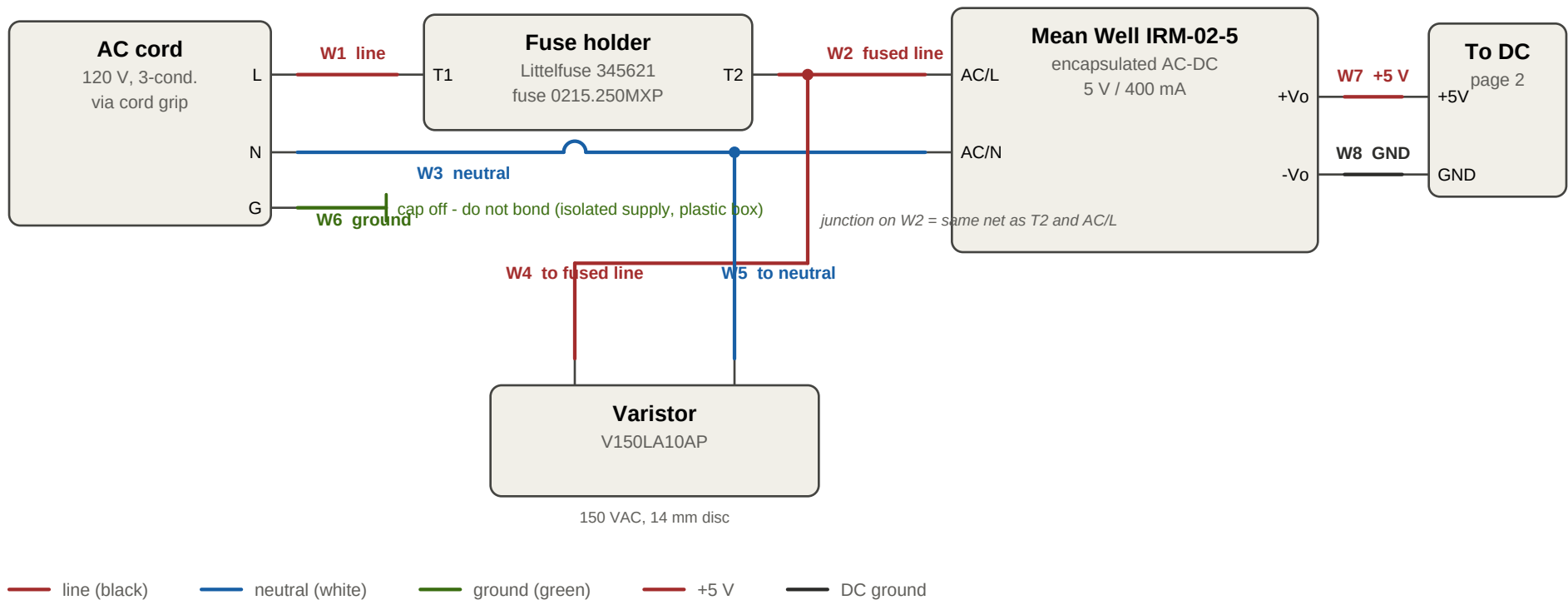


1. AC mains section

Cord entry, fuse, surge clamp and the Mean Well IRM-02-5. Every line is one physical wire.



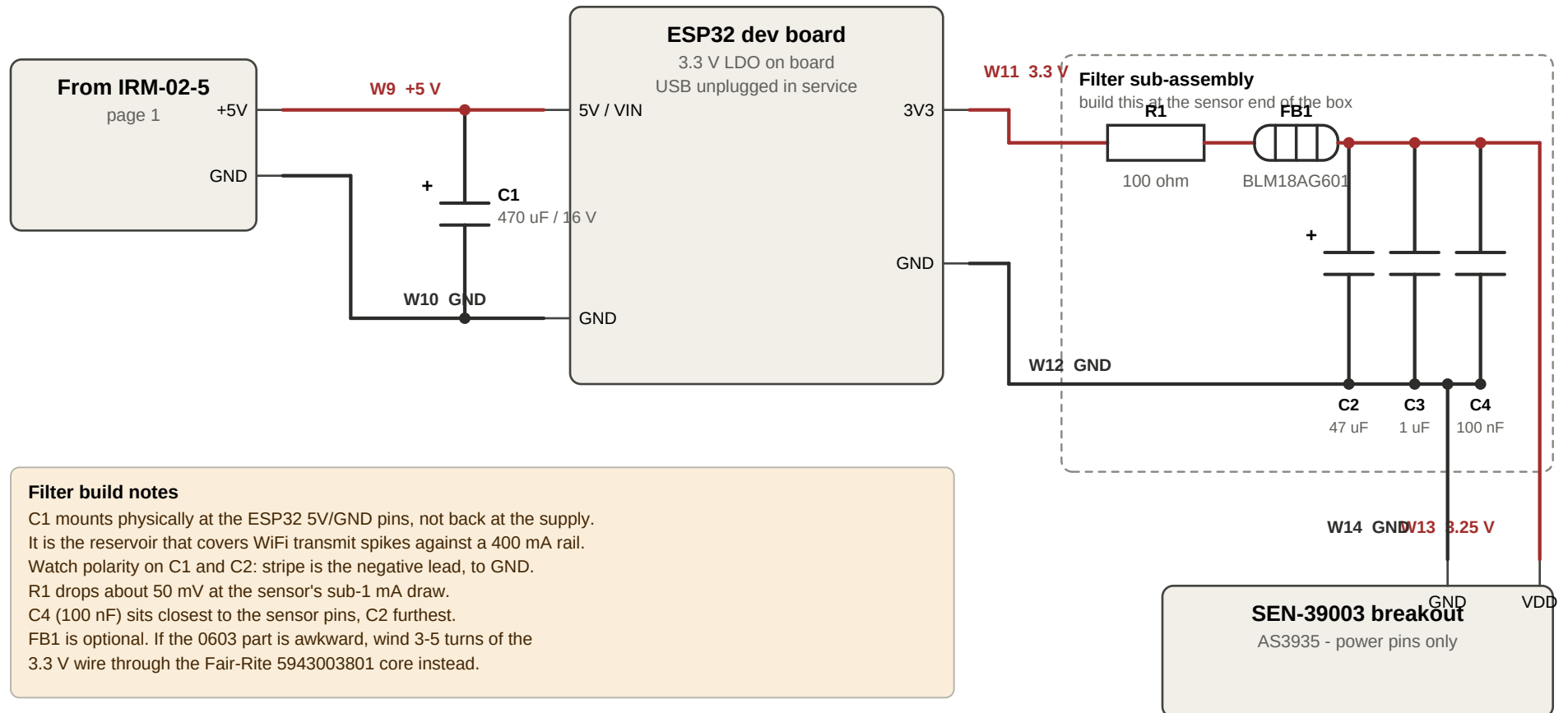
Mains rules

Fuse goes in the LINE conductor only, ahead of everything else.
MOV connects across line-neutral AFTER the fuse, so the fuse also protects you when the MOV fails short at end of life.
Sleeve every AC terminal. Keep 6 mm minimum from any DC wiring.
Mount the fuse holder through the box wall for outside access.

Isolation and ground

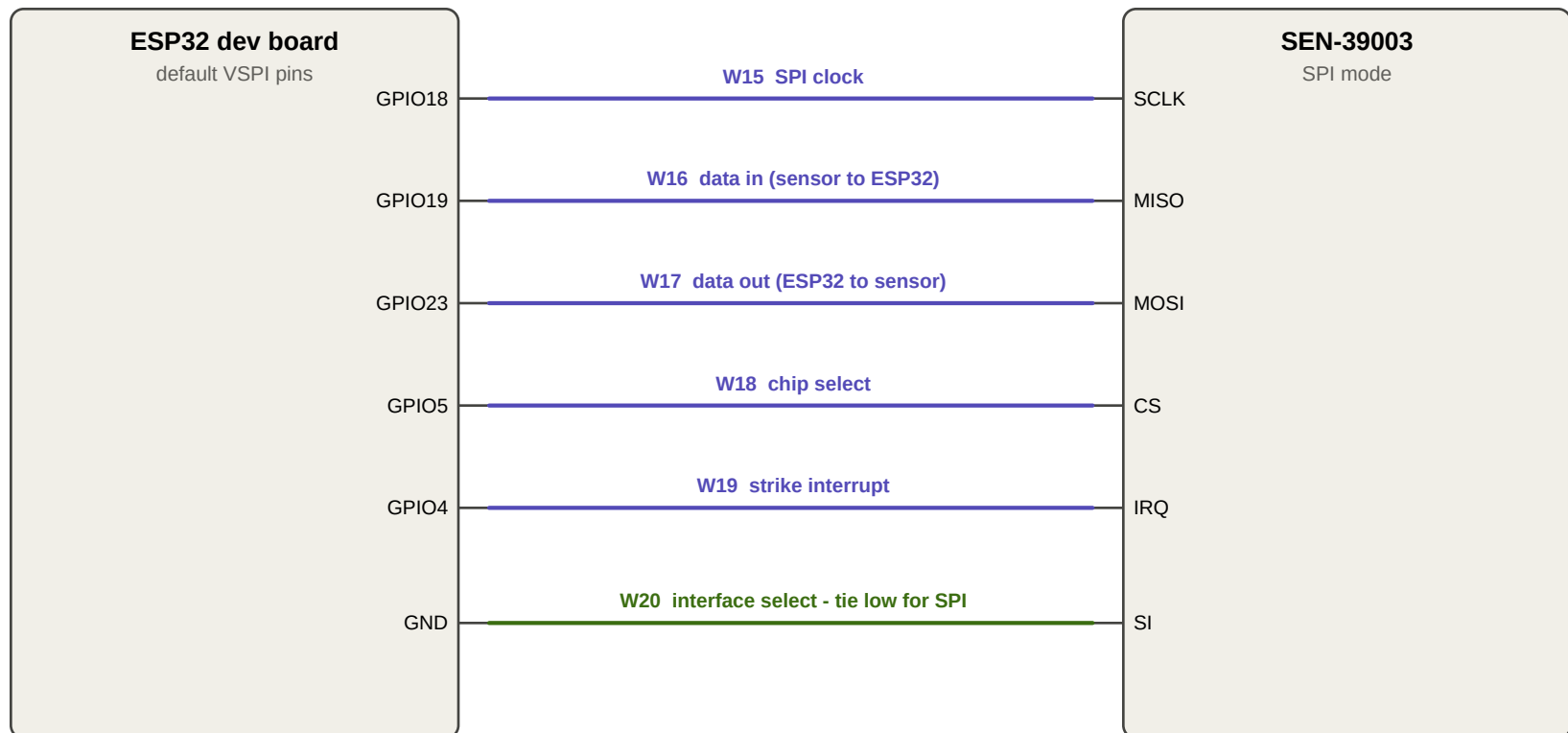
The IRM-02-5 output is galvanically isolated from the mains, so -Vo becomes your DC common. It is not earth.
Cord ground is left unterminated: the supply is 2-wire isolated and the enclosure is plastic, so there is nothing to bond it to.
Verify 5.0 V at +Vo / -Vo before connecting anything on page 2.

2. DC power and sensor rail filter



3. SPI and interrupt wiring

Six conductors between the ESP32 and the SEN-39003. Interface select must be tied low.



The one that catches people

SI selects the interface on this board: GND = SPI, VDD = I2C.
Leave SI floating and the sensor will not respond.
Do not use the Qwiic connectors - that path cannot carry the interrupt, which is the whole point of this build.

Routing

GPIO4 must stay a plain input - no pull-up conflict, no strapping pin. If you relocate it, avoid GPIO 0, 2, 12 and 15.
Keep this harness short (tens of cm) and route it away from the AC section and the supply. Twist nothing; just keep it tidy.
Signal ground is shared with the power ground on page 2.

4. Wire list and bring-up checklist

Tick each wire as you land it, then follow the power-up order.

	Wire	From	To	Note
☐	W1	AC cord - L	Fuse holder T1	line, black
☐	W2	Fuse holder T2	IRM-02-5 AC/L	fused line
☐	W3	AC cord - N	IRM-02-5 AC/N	neutral, white
☐	W4	MOV lead 1	Fused line node (W2)	after the fuse
☐	W5	MOV lead 2	Neutral node (W3)	
☐	W6	AC cord - G	unterminated	cap off, do not bond
☐	W7	IRM-02-5 +Vo	5 V rail	
☐	W8	IRM-02-5 -Vo	DC ground	becomes DC common
☐	W9	5 V rail	ESP32 5V / VIN	C1 across this and GND
☐	W10	DC ground	ESP32 GND	
☐	W11	ESP32 3V3	R1 lead 1	filter input
☐	--	R1 lead 2	FB1 lead 1	inside filter
☐	--	FB1 lead 2	Filter node	C2, C3, C4 land here
☐	W12	ESP32 GND	Filter ground bus	
☐	W13	Filter node	SEN-39003 VDD	approx 3.25 V
☐	W14	Filter ground bus	SEN-39003 GND	
☐	W15	ESP32 GPIO18	SEN-39003 SCLK	
☐	W16	ESP32 GPIO19	SEN-39003 MISO	
☐	W17	ESP32 GPIO23	SEN-39003 MOSI	
☐	W18	ESP32 GPIO5	SEN-39003 CS	
☐	W19	ESP32 GPIO4	SEN-39003 IRQ	per-strike interrupt
☐	W20	ESP32 GND	SEN-39003 SI	selects SPI

Bring-up order

1. Bench test the DC side on USB only, with the mains cord unplugged.
2. Confirm the sensor initialises and responds to the SEN-39002 emulator.
3. Unplug USB. Only then energise the AC side.
4. Measure 5.0 V at the ESP32 5V pin before landing W9.
5. Re-test the sensor on mains power. Disturbers that appear only now mean the filter needs more work, not a different mounting location.

Do not skip

Never have USB and the IRM-02-5 powered at the same time. On most dev boards the 5V pin ties straight to the USB rail and will back-feed your laptop.